MILITARY SPECIFICATION

MULTIMETER

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a general purpose non-electronic multimeter for measuring direct current (dc), dc voltage, alternating current (ac) voltage, and resistance.

2. APPLICABLE DOCUMENTS

* 2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116	Preservation, Methods of
MIL-E-17555	Electronic and Electrical Equipment Accessories, and Repair Parts, Packaging and Packing of

* STANDARDS

MILITARY

* MIL-STD-129	Marking for Shipment and Storage
* MIL-STD-794	Parts and Equipment, Procedures for Packaging of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: the Engineering Division, San Antonio ALC/MMEDO, Kelly AFB, TX 78241 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6625

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*(Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.1.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 Standard Practice for Commercial Packaging

(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

* 2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Qualification. The instrument furnished under this specification shall be a product which has been tested, passed the qualification tests specified herein, and has been listed on or approved for listing on the applicable qualified products list.

3.1.1 Sample articles furnished for qualification testing shall be manufactured by the prospective supplier requesting qualification and shall be constructed of the same type parts, components, subassemblies, and by the same manufacturing processes intended to be used during production. Sample articles shall have been tested in accordance with the qualification provision in 4.6, prior to submission for qualification testing.

3.1.2 Design changes occurring after qualification approval will require resubmission for qualification testing, unless a specific waiver is granted by the activity responsible for the qualified products list.

3.2 Design and construction. All circuitry shall exhibit sufficient stability of operating parameters to maintain calibration within the allowable tolerance without resetting any internal or external controls, for the environmental conditions, as specified in 4.7.1 and 4.7.2. For the shock and vibration tests specified in 4.7.3, external adjustment of the meter zero and the ohms zero shall be permitted before the individual tests are performed.

3.3 Functional description. This instrument shall be portable. It shall be calibrated to permit measurements of dc voltage from zero to 5,000 volts, ac voltage from zero to 5,000 volts, dc current from zero to 10 amperes, and resistance from zero to 10 megohms. All of these capabilities shall be internal functions of the multimeter. The accuracy of the multimeter shall be as follows:

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a. Bipolar dc voltage.

(1) ± 3 percent of full scale from zero to 1,000 volts.

(2) ± 5 percent of full scale from zero to 5,000 volts.

b. Bipolar dc current, ± 3 percent of full scale.

c. AC voltage.

(1) ± 4 percent of full scale from zero to 1,000 volts.

(2) ± 5 percent of full scale from zero to 5,000 volts.

d. Resistance, ± 3 percent of full scale length based on a linear scale.

3.4 Composition. This instrument shall consist of the following items and other component parts specified herein and as required to make up a complete instrument.

a. DC voltmeter section (3.5.1)

b. DC ammeter section (3.5.2)

c. AC voltmeter section (3.5.3)

d. Ohmmeter section (3.5.4)

e. Range selector switch (3.5.5)

f. Function selector switch (3.5.6)

g. Indicator section (3.5.7)

h. Overload protection circuit (3.5.8)

i. Accessories (3.5.9)

3.5 Performance requirements.

3.5.1 DC voltmeter section. The dc voltmeter shall measure dc voltages from zero to 5,000 volts in a minimum of 7 selective ranges. The lowest full scale reading shall not exceed 0.5 volt. The accuracy shall be within ± 3 percent of full scale from zero to 1,000 volts and ± 5 percent of full scale on the zero to 5,000 volts range. The sensitivity shall be at least 20,000 ohms per volt on all ranges with the exception of the lowest range. The sensitivity of the lowest range shall be at least 10,000 ohms per volt.

3.5.2 DC ammeter section. The dc ammeter section shall measure dc amperes from zero to 10 amperes in a minimum of 5 selective ranges. The lowest full scale reading shall not exceed 100 microamperes (ua). The accuracy shall be within ± 3 percent of full scale on all ranges.

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3.5.3 AC voltmeter section. The ac voltmeter section shall measure ac voltages from zero to 5,000 volts in a minimum of 6 selective ranges. The lowest full scale reading shall not exceed 3 volts. The accuracy shall be within ± 4 percent of full scale on all ranges from zero to 1,000 volts for a 60 cycle per second (cps) frequency ± 5 percent of full scale on the zero to 5,000 volts range for a 60 cps frequency. The sensitivity shall be at least 5,000 ohms per volt on all ranges.

3.5.3.1 The frequency response of this instrument shall be within ± 1 decibel (db) from 15 cps to 30 kilocycles (kc) on all ranges.

3.5.4 Ohmmeter section. The ohmmeter section shall measure resistance from zero to not less than 10 megohms in a minimum of 3 selective ranges. The accuracy shall be within ± 3 percent of full scale length based on a linear scale.

3.5.4.1 Ohms adjust control. An ohms adjust control shall be provided on the front panel of the instrument to adjust the meter to zero on the ohms range when the function selector switch and the range selector switch are set to ohms and ohmmeter leads are shorted.

3.5.5 Range selector switch. The range selector switch shall be located on the front panel of the instrument. This switch shall select the desired range of ac or dc voltage, dc current, or resistance.

3.5.6 Function selector switch. The function selector switch shall be located on the front panel of the instrument. This switch shall select the desired function of positive or negative dc voltage or current, ac voltage, or ohms.

3.5.6.1 The criteria in 3.5.5 and 3.5.6 may be provided in any desired combination of switches that furnish the stated requirements.

3.5.7 Indicator section.

3.5.7.1 Meter dial. The meter dial shall be of metal, finished with adherent, non-chipping, flat white enamel or lacquer. The dial scale markings shall be black and/or red for all scales. The ac scales shall be calibrated in root mean square (rms) volts and shall indicate the rms value of a sine wave. The length of the smallest scale shall be at least 3 inches.

3.5.7.2 Meter movement. There are no specific meter movement specifications. Satisfactory operation of the meter movement shall be determined by other tests specified herein.

3.5.7.3 Zero control. The meter shall have a control located on the front panel to mechanically zero the meter. When the meter is adjusted for zero deflection in the horizontal position, it shall also be zero when placed in the vertical position. The multimeter shall operate within instrument specifications in both the horizontal and vertical positions.

5.5.8 Overload protection.

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3.5.8.1 Overload relay circuit.

a. The overload relay circuit shall consist of a relay erated device and other necessary components. The relay shall energize id open the input circuit of the multimeter when the applied ac or dc oltage or current is at least 6 times rated full scale value. On the esistance range the overload relay will energize when the applied voltage s at least 6 times the voltage rating of the internal battery used on iat range. The multimeter shall not be damaged by the application of less ian 6 times rated full scale value.

b. A manual reset button shall be located on the front inel of the multimeter to reactivate the instrument for continued proper eration when an overload has been removed.

c. Proper operation of the overload relay circuit shall be idicated by the ability to zero the meter on the highest resistance range.

d. All full scale ranges, equal to or greater than 1,000 olts ac or dc, 1 ampere dc, or RX100,000 (if provided), shall be exempt om this overload relay requirement. The ranges that are not protected by ie overload relay shall be protected by appropriate fuses.

3.5.9 Accessories.

a. This instrument shall be provided with 2 test leads (one red and black). The test leads shall be constructed with banana or spring tip lug on one end and test prods together with insulated alligator clips on the ther end. The length of each test lead, including banana spring tip plug and est prod, shall be not less than 48 inches.

b. This instrument shall be provided with the batteries necessary or proper operation.

3.6 Electrical requirements.

3.6.1 Power supply. The power supply shall be contained internally and hall consist of a maximum of 2 commercial-type dry cells.

3.7 Mechanical requirements.

3.7.1 Case construction.

a. A detachable cover shall be provided for the case so that the nstrument face and panel are protected when the multimeter is not in use.

b. The case and cover shall preferably be of a nonmetallic aterial. This material shall be of sufficient strength to withstand the hock test as specified in 4.7.3.2.

3.7.2 Accessibility. The construction shall allow maximum accessibility internal parts and subassemblies.

3.7.3 Dimensions. The girth (face perimeter) plus the depth of the nstrument shall not exceed 35 inches.

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3.7.4 Weight. The unpacked weight of the instrument shall not exceed 7 pounds.

3.7.5 Controls. External controls shall be readily accessible, suitably arranged and of such size and construction to permit convenience and ease of operation under all service conditions.

* 3.7.6 Connectors. All input plugs shall be of standard banana spring tip type as specified in 3.5.9. The input plugs shall fit securely into the instrument's input jacks. A direct pulling force of at least 2 pounds shall be required to extract the input plugs from all instrument jacks and the selection or design of plugs and jacks used shall be capable of sustaining the required force during normal operation.

3.7.6.1 Recessing of input jacks. The top (exposed metal) of all input jacks shall be recessed at least 0.125 inch below the front panel.

3.7.7 Portability. The instrument shall be fitted with a carrying handle(s) capable of supporting the weight as specified in 3.7.4.

3.7.8 Markings.

* 3.7.8.1 Panel. Panel markings shall be stamped, etched or engraved and shall be filled with a permanent material which contrasts with the color of the instrument's panel.

3.7.8.2 Dials and switches. Dial and switch markings shall be stamped, etched or engraved and shall be filled with a permanent material which contrasts with the instrument's dials and switches when these markings are essential to the use and maintenance of the instrument.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. Examination, sampling and testing of the instrument shall consist of the following classifications:

a. Acceptance tests (4.3).

b. Qualification tests (4.6).

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4.3 Acceptance tests. Acceptance tests shall consist of:

a. Individual tests.

b. Sampling plan and tests.

4.3.1 Individual tests. After the instruments have been properly calibrated, each instrument shall be subjected to the individual tests specified in Table I, Group A. All of the items shall meet all of the requirements of the tests specified in Table I, Group A. The instruments shall meet all of the accuracies specified in 3.5 over the entire range and shall not be limited to the conditions stated in 4.5.

4.3.2 Sampling plan and test. One instrument, selected at random by a Government representative from each production lot of 50 or less serially numbered instruments produced on a contract or order, shall be subjected to all the tests specified in Table I, Group A and Group B.

Table I - Testing Index

Test	Requirement	Test Method
Group A: Individual Tests	3.1	4.3
a. Examination of product	3.2	4.5.1
b. DC voltage	3.5.1	4.5.2
c. DC current	3.5.2	4.5.3
d. AC voltage	3.5.3	4.5.4
e. Resistance	3.5.4	4.5.5
f. Indicator	3.5.7	4.5.6
g. Overload protection	3.5.8	4.5.7
*Group B: Environmental Tests		
a. Temperature, non-operating	3.2	4.7.1
b. Humidity	3.2	4.7.2
c. Vibration and shock	3.2	4.7.3
d. Mechanical fit of test lead plugs.	3.7.6	4.7.4

4.3.2.1 Delivery and acceptance. The production lot, from which the sample is selected, shall not be submitted to the Government for acceptance and delivery until the sample has satisfactorily completed all the tests specified in Table I, Group A and Group B.

4.3.2.2 Rejection and retest. When 1 instrument from a production lot fails to meet the specification, instruments still on hand or later produced shall not be accepted until the cause and extent of failures is determined and corrected. After corrections have been made, to the satisfaction of Government inspection personnel, all sampling tests shall be repeated. An additional sample of at least 3 instruments shall be selected from the same lot and shall be subjected to that portion of the test which produced the failure.

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4.3.2.3 Continuation of individual tests. Individual tests may be continued pending the investigation of a sampling test failure. Final acceptance of items on hand or later produced will not be made until the contractor determines (with Government concurrence) and demonstrates that the instruments meet all the requirements of this specification.

4.4 Equipment required. The following items of test equipment, or units of equal or greater accuracy and range, shall be used to perform the tests.

Equipment	Model
Variable Frequency Power Source	Empire Devices VP-1000
Instrument Standard Calibration Console	Weston 76
Audio Signal Generator	Hewlett-Packard 205AG
Test Oscillator	Hewlett-Packard 650A
Decade Resistor	General Radio 1432N
Decade Resistor	General Radio 1432P

4.5 Test method. The following tests shall be performed:

4.5.1 Inspection.

4.5.1.1 General. The multimeter shall be inspected for compliance to weight, overall dimensions, workmanship and general appearance.

4.5.1.2 Mechanical inspection. The multimeter shall be visually and mechanically inspected to assure compliance with all requirements of this specification. Particular attention shall be paid to workmanship, assembly and fit, materials, parts, freedom from binding, wiring assembly, solder joints, impregnation of coils, markings, freedom of parts from burrs and sharp edges, and accessibility of components for replacement.

4.5.1.3 Functional inspection. In the performance of all tests, the instrument shall be inspected to assure compliance with all the control and functional requirements set forth in 3 and 4 of this specification and to the specifications as stated by the manufacturer. Particular attention shall be paid to insure that the complete range of all control functions, within the tolerance, linearity and stability requirements, can be utilized and not necessarily be limited to the test points as specified in 4.

4.5.2 DC voltage.

4.5.2.1 Calibration accuracy.

a. Connect Weston Model 76 to a 115 volt, 60 cps source and allow 5 minutes for warm up.

b. Place multimeter in a horizontal position and zero the meter.

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c. From the Weston Model 76 voltage source, apply a positive full scale voltage to the multimeter. Repeat for each multimeter range from zero to 1,000 volts. To the 5,000 volts range, apply 2000 volts from the Weston 76.

d. The accuracy of the reading for each range from zero to 1,000 volts shall be within $\pm$ 3 percent of the applied voltage. When 2,000 volts is applied to the 5,000 volts range of the multimeter, the multimeter shall read within the limits of 1750 and 2250 volts.

e. Reverse output leads on the model 76 to give negative dc volts out. Set multimeter function switch to read negative dc volts.

f. Apply a negative full scale voltage to the multimeter. Repeat for each multimeter range from zero to 1,000 volts. To the 5,000 volt range, apply 2,000 volts from the Weston 76.

g. Repeat 4.5.2.1d.

h. Place multimeter in a vertical position and check meter zero.

i. Repeat 4.5.2.1c and 4.5.2.1d.

j. Repeat 4.5.2.1e through 4.5.2.1g.

4.5.3 DC current.

4.5.3.1 Calibration accuracy.

a. Set multimeter to measure positive dc current. Place multimeter in a horizontal position and check meter zero.

b. From the Weston Model 76, apply a positive full scale current to the multimeter. Repeat for each multimeter range.

c. The accuracy of the reading for each range shall be within $\pm$ 3 percent of the applied current.

d. Reverse output leads on the model 76 to give negative dc amperes out. Set multimeter function switch to read negative dc amperes.

e. Apply a negative full scale current to the multimeter. Repeat for each multimeter range.

f. The accuracy of the reading for each range shall be within $\pm$ 3 percent of the applied current.

g. Place multimeter in a vertical position and check meter zero.

h. Repeat 4.5.3.1b and 4.5.3.1c.

i. Repeat 4.5.3.1d through 4.5.3.1f.

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4.5.4 AC voltage.

4.5.4.1 Calibration accuracy (60 cps).

a. Set multimeter function switch to ac volts. Place multimeter in a horizontal position and check meter zero.

b. From the Weston 76, apply a full scale 60 cps voltage to the multimeter. Repeat for each multimeter range from zero to 1,000 volts. To the 5,000 volts range, apply 1,500 volts from the Weston 76.

c. The accuracy of the reading for each range from zero to 1,000 volts shall be within $\pm$ 4 percent of the applied voltage, when 1,500 volts is applied to the 5,000 volts range of the multimeter, the multimeter shall read within the limits of 1250 and 1750 volts.

d. Place multimeter in a vertical position and check meter zero.

e. Repeat 4.5.4.1b and 4.5.4.1c.

4.5.4.2 Frequency response.

a. Set multimeter function switch to ac volts and connect multimeter to frequency source.

b. Apply any 60 cps voltage to the multimeter. Vary the frequency from 15 cps to 30 kcs.

c. Selected voltage indication shall not change more than $\pm$ 1 db.

4.5.5 Resistance.

4.5.5.1 Calibration accuracy.

a. Set multimeter controls for measuring resistance. Short the test prods and adjust the multimeter for zero resistance by rotating the ohms adjust control. (Reposition the ohms adjust control for zero ohms each time the resistance range is changed.)

b. Use General Radio (GR) Model 1432N and GR Model 1432P (decade resistors) respectively for the low and high resistance ranges.

c. Connect the multimeter to appropriate decade resistor and apply a mid-scale resistance to each of the multimeter ranges. Note the multimeter reading on each range.

d. The accuracy of the reading for each range shall be within $\pm$ 3 percent of full scale length based on a linear scale.

4.5.6 Indicator. There are no direct indicator tests. Satisfactory operation of the indicator shall be determined by other tests specified herein.

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4.5.7 Overload relay. Set the range selector switch to highest resistance range and adjust the meter for zero ohms. This ability to zero the meter shall indicate proper operation of the overload relay circuit.

4.5.7.1 DC voltage overload.

a. Set multimeter to measure positive dc voltage.

b. Set Weston Model 76 for a dc voltage output of at least 6 times full scale voltage on the multimeter. Apply this voltage instantly to the multimeter. This shall cause the overload relay to deactivate the multimeter.

c. Remove the overload from the multimeter and depress the overload reset button.

d. Repeat 4.5.7.1b and 4.5.7.1c for each range of the multimeter (excluding the 1,000 and 5,000 volt ranges).

e. Set the Weston 76 for a dc voltage output of 4 times the full scale voltage on the multimeter. Apply this voltage instantly to the multimeter, then remove it.

f. Repeat 4.5.7.1e for each range of the multimeter (excluding the 1,000 and 5,000 volt ranges).

g. Repeat 4.5.7.1b through 4.5.7.1f with the Weston 76 supplying a 60 cps ac voltage instead of a dc voltage.

h. Apply a positive full scale dc voltage to any one of the ranges concerned in 4.5.7.1d. The accuracy of the multimeter shall be within specifications.

4.5.7.2 DC current overload.

a. Set multimeter to measure positive dc current.

b. Set Weston Model 76 for a dc current output of at least 6 times full scale current on the multimeter. Apply this current instantly to the multimeter. This shall cause the overload relay to deactivate the multimeter.

c. Remove the overload from the multimeter and depress the overload reset button.

d. Repeat 4.5.7.2b and 4.5.7.2c for each range of the multimeter (excluding the 1 ampere range and greater).

e. Set the Weston 76 for a dc current output of 4 times the full scale current on the multimeter. Apply the current instantly to the multimeter, then remove it.

f. Repeat 4.5.7.2e for each range of the multimeter (excluding the 1 ampere range and greater).

g. Repeat 4.5.7.2b through 4.5.7.2f with the Weston 76 supplying a 60 cps ac current instead of a dc current.

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h. Apply a positive full scale dc current to any one of the ranges concerned in 4.5.7.2d. The accuracy of the multimeter shall be within specifications.

4.5.7.3 AC voltage overload.

a. Set multimeter to measure ac voltage.

b. Set Weston Model 76 for a 60 cps ac voltage output of at least 6 times full scale voltage on the multimeter. Apply this voltage instantly to the multimeter. This shall cause the overload relay to deactivate the multimeter.

c. Remove the overload from the multimeter and depress the overload reset button.

d. Repeat 4.5.7.3b and 4.5.7.3c for each range of the multimeter (excluding the 1,000 and 5,000 volt ranges).

e. Set the Weston 76 for a 60 cps ac voltage output of 4 times the full scale voltage on the multimeter. Apply this voltage instantly to the multimeter, then remove it.

f. Repeat 4.5.7.3e for each range of the multimeter (excluding the 1,000 and 5,000 volt ranges).

g. Repeat 4.5.7.3b through 4.5.7.3f with the Weston 76 supplying a positive dc voltage instead of a 60 cps ac voltage.

h. Apply a full scale 60 cps ac voltage to any one of the ranges concerned in 4.5.7.3d. The accuracy of the multimeter shall be within specifications.

* 4.5.7.4 Resistance overload.

a. Set multimeter to measure resistance.

b. Set Weston Model 76 for a dc voltage output of at least 6 times the voltage rating of the internal battery used on that range. Apply this voltage to the multimeter. This shall cause the overload relay to deactivate the multimeter.

c. Remove the voltage from the multimeter and depress the overload reset button.

d. Repeat b and c for each resistance range of the multimeter (excluding RX100,000 if provided).

4.6 Qualification testing.

4.6.1 Qualification requirements. Prior to actual contract award, all products submitted by any manufacturer shall have passed the qualification tests as specified in Table I, Group A and Group B.

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4.6.2 Qualification test samples. The qualification test samples shall consist of 3 instruments complete with accessories as specified herein. Samples shall be true production articles selected from the latest production lot and shall be identified by the manufacturer's own type or model number. The manufacturer shall submit a complete alignment and calibration procedure with each sample. The tests, as sequenced in this specification, do not constitute proper calibration procedures. Qualification samples shall be accompanied by a certification that these samples have been subjected to and have passed the tests as specified herein. Records of the inspections and tests for each unit supplied shall be kept complete and shall be made available to the Government upon request.

4.6.3 Discontinuance of tests. The Government maintains the prerogative of stopping the qualification testing at any time the test article fails to meet the specification requirements. Return of the test sample to the manufacturer may be required whenever the Government determines the test article has not complied with the specifications requirements.

4.6.3.1 Failure. Random component failures shall not necessarily constitute grounds for disqualification. Repeat failures shall be considered grounds for disqualification. Repeat failures are defined as follows:

a. More than 1 failure of a specific component (reference symbol) in a single test article.

b. Failure of the same component (reference symbol) in more than 1 sample article.

c. More than 1 failure of different but related components in a circuit.

4.6.3.2 Adjustments during qualification testing. After the qualification tests have started, only adjustments to externally available controls, without the aid of tools, will be permitted. Any other corrections shall be considered a failure.

* 4.6.4 Resubmission. Samples shall not be resubmitted within six months after a failure to qualify, unless specifically waived by the responsible activity. Samples may then be resubmitted for qualification testing upon receipt and evaluation by the Government of documented evidence of corrective action. A certified test report, in accordance with the specification requirements, for all samples to be resubmitted for qualification testing shall be delivered to the Government, prior to resubmission of test articles. These test reports shall include all the tests performed and the test results.

4.7 Test method, group B. The following tests shall be performed in accordance with 4.3 and 4.5.

4.7.1 Temperature test, non-operating.

4.7.1.1 Low temperature test. Place the instrument in a temperature test chamber and reduce the temperature of the chamber to minus 54°C. Maintain this temperature for 4 hours. Return the test chamber and instrument to room temperature in 30 minutes. Remove the instrument from the test chamber. After the instrument has stabilized to room temperature, repeat all of the individual tests specified in 4.5. The instrument shall perform within the specified tolerances and there shall be no indication of failure as a result of this test.

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4.7.1.2 High temperature test. Place the instrument in a temperature test chamber and elevate the temperature of the chamber to plus 80° C. Maintain this temperature for 4 hours. Return the test chamber and instrument to room temperature in 30 minutes. Remove the instrument from the test chamber. After the instrument has stabilized to room temperature, repeat all of the individual tests specified in 4.5. The instrument shall perform within the specified tolerances and there shall be no indication of failure as a result of this test.

4.7.2 Humidity test cycle. Place the multimeter in a humidity test chamber and take the chamber to 95 percent humidity at 38°C. Maintain this humidity-temperature relationship for 24 hours. Reduce test chamber to 81 percent humidity at 22°C in a period of 2 hours and remove multimeter from chamber. From the Weston Model 76, apply a positive full scale voltage of 1,000 volts dc to the multimeter and note the multimeter reading. The accuracy of the reading shall be within ± 3 percent of the applied voltage.

4.7.3 Vibration and shock.

* 4.7.3.1 Vibration test. The instrument shall be rigidly attached, in the normal operating position, to a vibration testing mechanism. The instrument shall be subjected to a total displacement of 0.01 inches at 10 cps and 50 cps for 5 minutes at each frequency. The total displacement shall be vertical (right angles) with respect to the normal operating position of the instrument. External adjustment of the meter zero and ohms zero shall be accomplished after the completion of this vibration test. After making these adjustments perform the individual tests as specified in Table I, Group A. The instrument shall operate within the specified tolerances. There shall be no evidence of physical damage or circuitry failure as a result of this test.

4.7.3.2 Shock. The multimeter shall be held a minimum of 20 inches from a concrete floor (or slab) and dropped once from a horizontal position so as to land on its back. The multimeter (after adjusting the meter zero and ohms zero) shall operate within the specified tolerances. There shall be no evidence of physical damage or circuitry failure as a result of this test.

* 4.7.4 Mechanical fit of test lead plugs. Insert each input plug fully into each of the instrument's input jacks, then extract the plug. Repeat this procedure at least 25 times. On each fifth extraction, measure the pulling force required to extract the plug from the jack. The extraction force shall be at least 2 pounds on each pull, and there shall be no evidence of physical damage or degradation in the extraction force as a result of this test.

* 4.7.4.1 Recessing of input jacks. Measure and record the distance from the top of the front panel to the top of the exposed metal on each input jack. The measured distance shall be at least 0.125 inch.

* 5. PACKAGING

* 5.1 Preservation shall be Level A, C or Commercial/Industrial as specified.

* 5.1.1 Level A. The multimeters shall be preserved and packaged in accordance with the applicable Level A requirements of MIL-E-17555 and MIL-P-116.

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* 5.1.2 Level C. Level C. The multimeters shall be preserved and packaged in accordance with the applicable Level C requirements of MIL-E-17555 and MIL-P-116.

* 5.1.3 Commercial/Industrial. The commercial preservation of the multimeters shall be in accordance with the requirements of ASTM D 3951.

* 5.2 Packing. Packing shall be Level A, B, C or commercial/industrial as specified.

* 5.2.1 Level A. The packaged multimeters shall be packed in containers as specified in MIL-STD-794 for Level A protection.

* 5.2.2 Level B. The packaged multimeters shall be packed in containers as specified in MIL-STD-794 for Level B protection.

* 5.2.3 Level C. The packaged multimeters shall be packed in containers as specified in MIL-STD-794 for Level C protection.

5.2.4 Commercial/Industrial. The packaged multimeters shall be packed in accordance with the requirements of ASTM D 3951.

5.3 Marking.

5.3.1 Levels A, B and C. In addition to any special or other identication marking required by the contract each unit pack, intermediate and exterior container and unitized load shall be marked in accordance with MIL-STD-129.

5.3.2 Commercial. Commercial/industrial marking shall be in accordance with the requirements of ASTM D 3951.

6. NOTES.

6.1 Intended use. The multimeters covered by this specification are intended for use in environmental control settings.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Levels of preservation-packaging and packing (see 5.1).

6.3 Qualification: With respect to products requiring qualification, awards will be made only for products which are, at the time set for opening of bids, qualified for inclusion to Qualified Products List (QPL No.) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible

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to be awarded contracts or purchase orders for the products covered by this specification. The activity responsible for the Qualified Products List is San Antonio ALC/MMEDO, Kelly AFB, TX 78241 and information pertaining to qualification of products may be obtained from that activity.

6.4 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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MIL-M-38706	MULTIMETER

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	☐ OTHER *(Specify)* _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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